Cycle 01 — Results & learning

CYCLE 01: FROM QUESTION TO MEASUREMENT

Project Gravity Witness • 17 October 2026 • First research results

Start here: the five-minute briefing

The program has begun. We now have a checked two-mass calculation, a concrete three-setting entanglement readout, and implementations of two very different comparison models. One cannot entangle the masses in the modeled regime; the other can. The task is now sharper: determine which alternatives survive real physical constraints, then look for measurement records that separate them.

STATUS: REPRODUCTION AND PRELIMINARY DESIGN. No new physical discovery, apparatus demonstration, complete discriminator against all viable alternatives, external physics review, or Lean-kernel verification is claimed. The 970 passing checks are numerical and software checks, not 970 proofs.

A correction matters. The initial charter treated the Aziz–Howl classical-gravity entanglement claim too confidently. Direct rebuttals contest its dynamics and/or attribute the effect to quantum matter exchange. The conditional local-mediator witness has not simply been overturned. We will keep the prediction, its assumptions and its physical interpretation separate. [S04, S14, S15, S19, S22]

The most useful result for your oversight is that the theory now ends in a measurable score, not just an impressive equation. In an explicitly idealized example, that score is 1.200 versus a separable-state ceiling of 1. The assumed noise performance has not been achieved or validated; it tells us what the experiment would need to preserve.

Read the next three sections to understand that score. Then read the harder countermodel and the two traps. The full derivations, source audit, failed checks and code are in the private archive linked at the end.

Lesson 1 — What is changing in the four paths?

Imagine two quantum wave patterns, each with a left and a right branch. There are four joint alternatives: LL, LR, RL and RR. Gravity changes the angle, or phase, of each alternative. Phase is like a hand on a clock: a common rotation changes no relative reading, but a difference between clocks changes their relationship. The analogy describes phase only; ordinary clocks are not entangled quantum systems.

The joint phase is $\Delta = \varphi_{LL} + \varphi_{RR} - \varphi_{LR} - \varphi_{RL}$. A phase that belongs only to A, only to B, or to all four alternatives cancels. In the ideal equally weighted state, concurrence $C = |\sin(\Delta/2)|$.

Here C is an entanglement measure from 0 to 1. Negativity N is another measure, from 0 to 0.5 for two qubits; in this pure-state example N = C/2. [S01, S16–S18; reproduced in gw.py]

Worked example: assume two point masses of 10^{-14} kg, centre separation d = 450 micrometres, branch splitting s = 250 micrometres, and a 2.5-second static hold. The four distances are 450, 700, 200 and 450 micrometres. The calculation gives $\Delta = -0.313934$ radians, C = 0.156323 and N = 0.078162. These are illustrative inputs, not an experimental achievement or a full interferometer simulation.

A useful surprise: all four path populations remain 25%. Counting which path each mass occupies would miss this entanglement. We must recombine or rotate the quantum states and measure the right correlations.

Lesson 2 — The actual measurement score

Use fresh prepared pairs for three measurement settings: XX, YY and ZZ, after calibrated local rotations. X, Y and Z mean perpendicular spin measurement axes. Each spin measurement returns +1 or −1. A correlation such as $\langle XX \rangle$ is the average product of the two outcomes in that setting.

Compute $S = \langle XX \rangle - \langle YY \rangle + \langle ZZ \rangle$. Every separable two-qubit state satisfies $S \leq 1$ under these trusted-measurement assumptions. A separable state is a mixture of independent states; it may still have ordinary correlations. The bound follows from the geometry of the two local spin vectors and survives mixing. This is a standard entanglement witness in suitable local frames, not a new Bell test or a device-independent proof.

For our example, assume independent dephasing retains $\lambda = 0.95$ of each local branch coherence. Then S = 1.199515 and N = 0.049879. The code supplies all four outcome probabilities for each setting. In this particular state/noise family, $S = \lambda^2 + 2\lambda|\sin(\Delta/2)|$. The threshold is $\lambda = 0.855821$, not a universal coherence requirement.

An observed S > 1 would establish entanglement only after accounting for statistical and measurement errors. It would not establish that gravity caused it. Initial entanglement, electromagnetic coupling, quantum matter exchange, unwanted selection of outcomes, and spin/readout artifacts must be independently excluded or bounded.

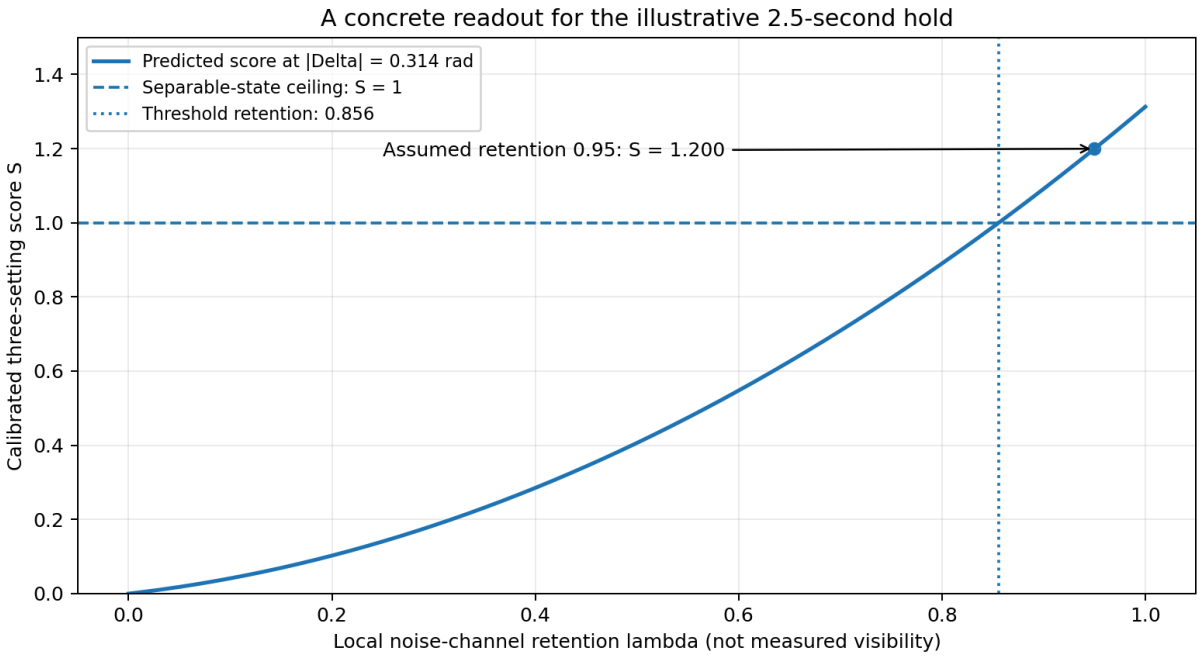


Figure C01-A. Calculated readout versus assumed noise-channel retention for the 2.5-second example. The point at 0.95 is an assumption, not measured performance. Generated by run_cycle01.py; claim GW-C003.

What would count as evidence?

A preliminary protocol is now specified: prepare independently, allow an interaction for a prescribed time, apply calibrated local rotations, collect the three joint-outcome distributions, and compare the score with its separable bound. Repeat over preselected times and geometries while collecting matched controls. No postselection or favorable parameter changes after looking at the result.

The archive includes a conservative sampling example: 3,968 prepared pairs per setting would suffice under its stated independent-trial, fixed-sample and bounded-bias assumptions. This is not a prediction of laboratory time or feasibility. A real noise budget and a guaranteed implemented signal are missing. We will not convert a toy statistical bound into an apparatus promise.

Lesson 3 — An easy opponent and a harder one

The Kafri–Taylor–Milburn model represents gravity through measurement and classical feedback. Its source equation couples the force to enough decoherence to prevent entanglement in our static two-branch restriction. We reproduce that behavior analytically and numerically. It is a useful benchmark, not the strongest currently viable alternative; particular pairwise extensions already face experimental constraints. [S02, S21]

The harder comparison is the Gaussian-smeared Diósi–Penrose model studied by Trillo and Navascués. We implemented its correlated noise, not a convenient independent-noise

substitute. The implementation reproduces published limiting distance ratios of 0.850872 and 2.210928, and a specific example that DOES entangle. A separate Lindblad propagator agrees with the direct solution. [S05; dp.py and test_dp.py]

This does not prove a viable classical-gravity loophole. The interpretation of the model as local classical mediation is disputed, and our example has not passed real-world constraints. The cutoff length σ, finite size of the objects, shielding, heating and force-law limits must be audited before calling it a viable rival. A 2026 paper by Aspelmeyer argues that actual table-top constraints change the collapse-model conclusion; its quantitative argument is a priority for the next cycle. [S08, S20]

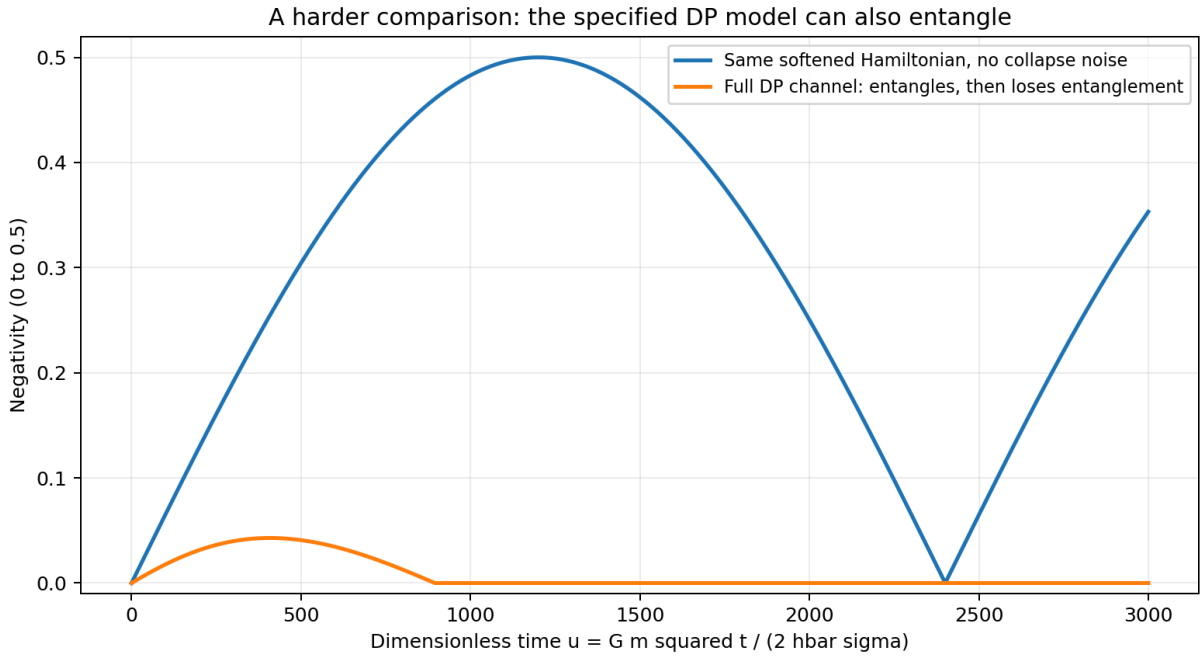


Figure C01-B. The specified DP dynamics can entangle, but not like a noiseless evolution with the SAME softened gravitational Hamiltonian. Time is dimensionless, $u = Gm^2t/(2\hbar\sigma)$. This is not a comparison with point-Newton gravity at unmatched coupling. Generated by run_dp.py; claim GW-C006.

Two traps the research must avoid

TRAP 1: Inconsistent approximations. The frequently useful small-splitting approximation gives a joint phase smaller than the exact point-Newton result by the fraction $(s/d)^2$. Our illustrative 250/450 geometry therefore has a 30.9% approximation error. Comparing an exact quantum calculation with an approximate classical one could manufacture an apparent difference. At s/d = 0.1, the error is 1%. This is an elementary guardrail we derived, not a new discovery or an error attributed to the original papers.

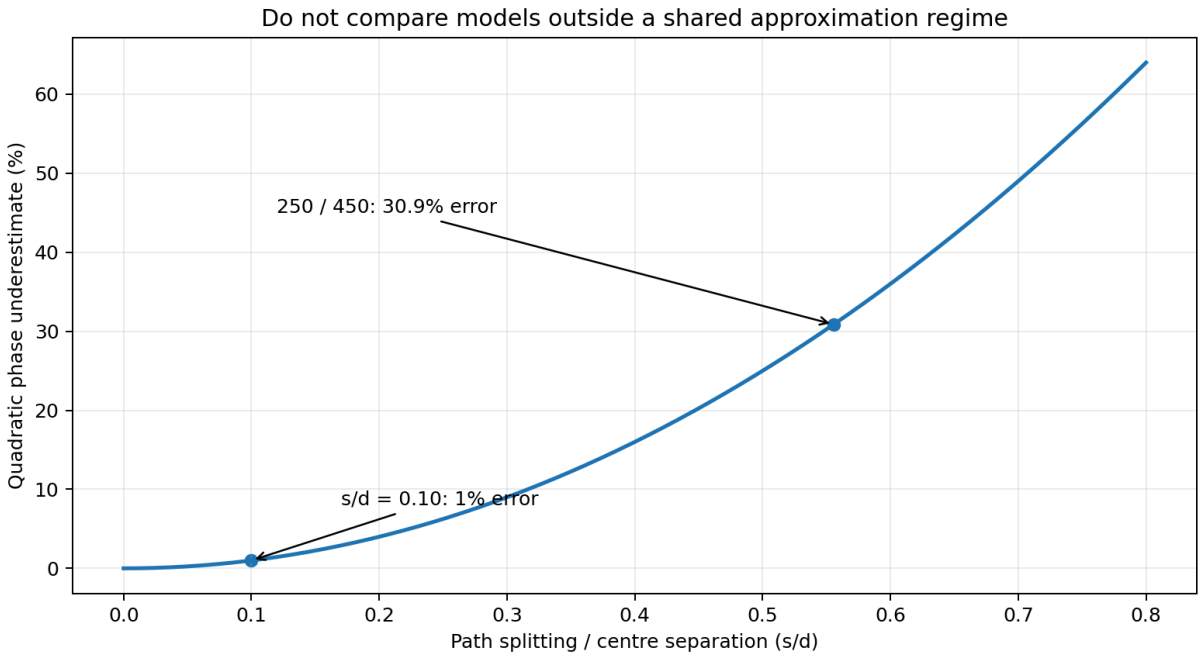


Figure C01-C. Exact relative error of the quadratic inline phase approximation. A fair comparison needs a shared accuracy regime or consistently exact models. Generated by run_cycle01.py; claim GW-C004.

TRAP 2: Mistaking lost interference for environmental noise. For the stated family, final single-particle visibility is V = λ|cos(Δ/2)|, not λ. At Δ = π it vanishes even with no environmental decoherence, because the masses are entangled. A separate matched control is needed to identify noise retention. That control's physical validity must also be tested. [Derived and numerically cross-checked; GW-C011]

What has actually been checked?

970 parameterized tests pass with zero failures, errors or skips. Checks cover pure-state identities, local-phase invariance, partial-transpose eigenvalues, unequal dephasing, separable negative controls, independent matrix-exponential implementations, density-matrix validity, and selected DP numerical limits. SymPy checks selected exact algebra; a 60-digit scalar calculation checks a DP onset value. These are cross-checks within this project, not external expert replication.

The failure log retains a symbolic-library mismatch, a comparison against a paper's approximate printed number, approximation hazards and access limits. A Lean draft exists for narrow determinant and phase identities, but Lean and mathlib could not be installed here. None of those draft statements is labeled kernel-verified.

The current answer to our research question is still conditional: we have a concrete readout and checked model comparisons, but not yet a realizable protocol that robustly excludes the strongest viable alternatives. “Viable” must be earned by an independent constraint audit, not granted because a model produces a graph.

Your next review gate

First, determine whether the harder DP example survives finite-body, cutoff, heating, force-law and shielding constraints. Next, compare full measurement distributions from surviving models in the SAME apparatus geometry, with bounded nuisance channels. Keep the gravitational-noise/coherence and optical cross-correlation tracks open; we have not selected the two-mass route as the ultimate best experiment.

For your own check: Why do four equal path counts not rule out entanglement? Why can a score above 1 establish entanglement without identifying gravity as its cause? Why is the DP model's mathematically entangling example not automatically an experimentally viable counterexample? Why is final visibility not a direct measure of environmental noise?

Code, provenance and continuity

The complete research package contains 37 tracked files: Python models, tests, five generated figures, numerical outputs, the source registry, claim ledger, explanatory notes, failure log and uncompiled Lean draft. The local Git commit is ea86abd353217f9c65a8b36a1945573ff8f1a796. The archive includes the Git history bundle and a SHA-256 manifest for its tracked files.

Private research archive: gravity-witness-cycle01.zip

The archive is saved privately in Drive. GitHub remote creation remains blocked because the connected action set has no create-repository operation; no remote push is claimed. The intended private repository remains dhinchcliffe/gravity-witness. No unattended executor or recurring research schedule was launched in this cycle.

Primary sources for this cycle

Source IDs retain the charter's numbering. The archive's sources.json records what was actually read or reproduced; a listed paper is not automatically an audited result.

S01 — Bose et al., 2017: branch-phase experiment framework.

S02 — Kafri, Taylor and Milburn, 2014: classical-channel benchmark.

S05 — Trillo and Navascués, 2025, arXiv v2: selected DP equations reproduced.

S08 — Feng, Vedral and Marletto, 2026: locality challenge to collapse-model interpretation.

S14 — Gundhi, Infantino and Bassi, 2026: direct challenge to Aziz–Howl; version-access limits recorded.

S15 — Tibau Vidal and Iyer, 2026: matter-exchange and subsystem interpretation.

S16–S18 — Wootters; Horodecki et al.; Peres: concurrence and partial-transpose criteria. Full links in sources.json.

S19 — Marletto and Vedral, 2017: conditional mediator witness.

S20 — Aspelmeyer, 2026: actual table-top constraints; quantitative reproduction still pending.

S21 — Altamirano et al.: constraints on a pairwise local classical channel.

S22 — Marletto, Oppenheim, Vedral and Wilson, 2025: direct rebuttal to the classical-mediator interpretation.

Change record

2026-10-17 | v0.2. Charter approved; Cycle 01 executed. Added checked calculations and illustrated learning notes, preserved the charter and original diagram, corrected the overstated classical-gravity claim, and recorded infrastructure and scientific limits. No novelty or experimental-success claim.

Charter & foundations

PROJECT GRAVITY WITNESS

An explanatory research atlas and experimental-discriminator program

Research atlas v0.2 • 17 October 2026 • APPROVED AND STARTED — FIRST REPRODUCTION CYCLE COMPLETE. Begin with the Cycle 01 tab.

READ THIS FIRST

The question in one minute

Gravity, as Einstein describes it, is the geometry of spacetime. Quantum mechanics describes particles and fields with superposition, interference and entanglement. Both theories work spectacularly in their domains. We do not yet know whether gravity itself must obey quantum rules. An experiment can help by comparing precisely specified models, rather than attempting to prove every imaginable theory wrong.

The motivating experiment is almost childlike in its structure: prepare each of two tiny masses so that it can follow a LEFT or a RIGHT path in quantum superposition. Let the masses interact while carefully suppressing non-gravitational forces. Then recombine the paths and measure correlations. Four path combinations experience four different gravitational distances. Their quantum phases can encode entanglement.

CORRECTION / DISPUTED: Some proposed classical or hybrid models predict entanglement, but their locality assumptions, physical viability and interpretation are contested. In particular, the Aziz–Howl claim has direct technical rebuttals. The conditional local-mediator witness has not simply been overturned. Our task is to compare precisely specified predictions while auditing initial separability, other quantum channels, subsystem definitions and experimental constraints. See the Cycle 01 tab and sources S04–S08, S14, S15, S19, S20 and S22.

The north-star question

What realizable experimental protocol—or coordinated set of measurements—would distinguish the predictions of low-energy quantized gravity from strong, explicitly defined classical-gravity/quantum-matter alternatives, after realistic noise, confounders and theoretical assumptions are included?

Our result is not a claim to have unified physics. The target is a testable discriminator, with verified derivations, identified loopholes, parameter ranges and honest limits of inference.

How to read this atlas

Every major concept will have five layers: (1) 60-second intuition; (2) everyday analogy; (3) precise definition; (4) a worked numerical or mathematical example; (5) a “what would disprove this?” challenge. Every displayed claim will be labeled ESTABLISHED, REPRODUCED,

PROPOSED, DISPUTED, or OPEN. New mathematics must have a linked code result, analytic derivation or independently checked proof.

Figure 1 — the thought experiment

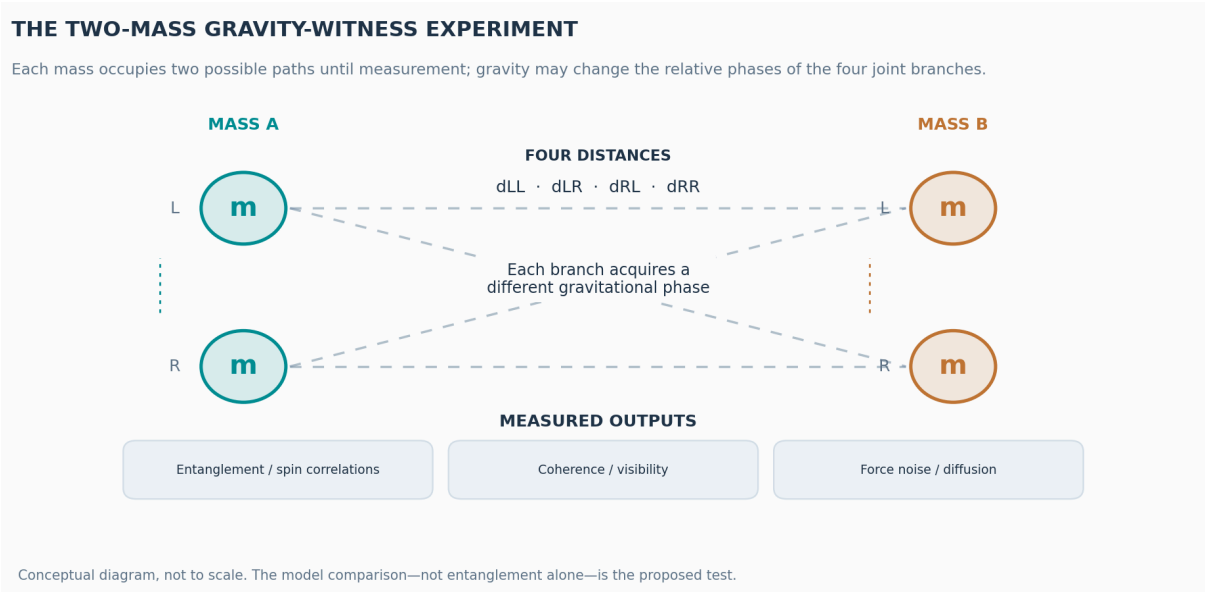


Figure 1. A conceptual—not-to-scale—view of the four branch distances and three kinds of observables. Initial versions of this document will use original, reproducible figures; every subsequent major result gets a diagram with units and a link to its generating code.

THE ESSENTIAL VOCABULARY

Classical gravitational field. A field described by ordinary numbers (for example a spacetime metric), rather than a quantum superposition of field states. Classical does NOT necessarily mean noiseless, deterministic, or compatible with every intuitive notion of locality.
Quantum gravitational field. A gravitational field with genuinely quantum degrees of freedom; the weak-field, low-energy baseline can be treated in perturbative quantum theory without requiring a complete theory of the Planck regime.
Superposition. A quantum state with amplitudes for multiple alternatives, such as LEFT and RIGHT, capable of interfering. This is not a mass literally chopped into two pieces.
Entanglement. Two quantum systems are entangled if their shared state cannot be expressed as a separable combination of independent states. A correlation alone is not enough: classically correlated objects can share information without entanglement.
Phase. The angle of a complex quantum amplitude. Relative phase differences change interference and can create entanglement when they depend jointly on both masses' positions.
Coherence / decoherence. Coherence is the preservation of relative quantum phases; decoherence is their loss through interaction or averaging over uncontrolled degrees of freedom. Reduced visibility can arise for many non-gravitational reasons.

Interferometer. An apparatus that splits and recombines alternative paths, turning phase changes into measurable differences. A Stern–Gerlach apparatus can couple a spin state to alternative paths; a Michelson interferometer uses optical paths.
Mediator. The mechanism that transmits an interaction between systems. “Gravity creates entanglement” and “the gravitational FIELD must be quantum” are different claims unless the mediator's allowed operations are specified.
LOCC / locality. Local operations and classical communication cannot create entanglement from an initially separable state. However, the information-theoretic sense of locality in this theorem is not identical to spatiotemporal locality. This distinction is central to the 2025–2026 disagreement. [S06–S07]
Backreaction. The effect of quantum matter on the gravitational field that in turn acts on matter. A viable hybrid theory must specify this interaction, not only gravity acting on a prescribed quantum system.
Stochasticity / diffusion. Random behavior governed by probabilities, including possible fluctuations of spacetime or the motion of matter. An otherwise-classical gravity theory may imply extra diffusion or decoherence.
Discriminator. An observable or combination of observables for which competing models predict measurably different distributions after uncertainties and nuisance parameters are considered.

THE CANONICAL TWO-MASS EXAMPLE

Intuition: four possible worlds, one shared experiment

Mass A can be at L or R, and mass B can be at L or R. The two-mass system has four joint alternatives: LL, LR, RL and RR. Each pair has its own separation d_{ij}. Gravity acts differently at different distances, so each alternative acquires a different phase over a duration t. After recombination, quantum spin correlations can reveal whether the two systems were entangled.

A small piece of mathematics

Under an idealized Newtonian weak-field model with sharply localized branches and equal masses m, an interaction-dependent branch phase has magnitude $\varphi_{ij} = G m^2 t / (\hbar d_{ij})$. Here G is Newton’s gravitational constant, $\hbar$ is the reduced Planck constant, t is interaction time and d_{ij} is branch separation. This expression by itself is a toy-model phase, not a complete experimental prediction. [S01, S04]
The part that matters for two-particle entanglement is $\Delta\varphi = \varphi_{LL} + \varphi_{RR} - \varphi_{LR} - \varphi_{RL}$. In an ideal pure, equally weighted two-qubit state with only branch-dependent phases, the concurrence is $C = |\sin(\Delta\varphi/2)|$. C = 0 means no entanglement; C = 1 is maximal entanglement. A common phase that affects all four branches equally cancels. (Our first Lean target is the precise finite-dimensional algebra behind this claim.)
Worked example: if $\Delta\varphi = 0$, there is no entanglement in this toy state. If $\Delta\varphi = \pi$, C = 1. In a realistic apparatus, the branches are wave packets rather than mathematical points;

spin-readout visibility, decoherence, thermal motion and unintended electromagnetic coupling complicate the interpretation.
First challenge question: Could an apparent correlation come from charge, magnetic fields or shared preparation noise rather than gravity? Second: If a classical-gravity model produces nonzero C through quantum matter, what else can we measure to tell it apart?

COMPETING THEORIES — FIRST MODEL REGISTRY

M0 | Quantized weak-field gravity (baseline). A low-energy quantum interaction model predicts branch-dependent phases and potentially gravity-mediated entanglement in appropriate preparations. This is NOT a complete ultraviolet theory of quantum gravity. Track all approximations.
M1 | Classical measurement-and-feedback channel (Kafri–Taylor–Milburn). Gravity is represented as classical communication, with associated decoherence and heating. Such a channel cannot create entanglement from an initially separable input under its assumptions. Test its model-specific noise tradeoffs. [S02]
M2 | Mean-field semiclassical gravity / Schrödinger–Newton. Quantum matter interacts with a field sourced by an expectation value. State-dependent dynamics may produce optical cross-correlations distinguishable in a specially designed interferometer. Test that specific model rather than all classical gravity. [S10]
M3 | Stochastic classical-quantum gravity (Oppenheim and related work). Classical spacetime couples to quantum matter through probabilistic dynamics. Predicts characteristic relationships involving decoherence, metric fluctuations and diffusion; realizations and consistency constraints matter. [S03, S09]
M4 | Diósi–Penrose-type collapse dynamics. REPRODUCED IN A LIMITED MODEL: Cycle 01 independently implements selected static Gaussian-smeared dynamics and reproduces an entangling example. This does not establish local classical mediation or experimental viability. Locality and actual table-top constraints require separate audits. Preserve the predictions and the objections. [S05, S08, S20]
M5 | Classical gravitational field with QFT matter (Aziz–Howl). DISPUTED, not an established viable counterexample. The Nature paper claims entanglement with classical gravity; direct rebuttals challenge the calculation and/or identify quantum matter exchange rather than gravity as its mediator. Full QFT calculations have not been reproduced in this project. Audit subsystem partition, retained amplitudes, matter propagation and postselection before using this model. [S04, S14, S15, S22]
MODEL AUDIT RULE. For each model record the exact state space and dynamics, fields treated as classical/quantum, assumptions about locality, complete positivity, no-signaling and conservation, adjustable parameters, empirical constraints, and explicit predictions for every measurable quantity. Do not call a model viable merely because it appears in a paper.

THREE EXPERIMENT FAMILIES — THE INITIAL BAKE-OFF

Candidate A | Two masses, one gravitational interaction, several readouts

Preparation: independent spatial superpositions of two small masses. Readouts: spin correlations or another entanglement witness, single-particle coherence/visibility, heating and environmental noise. Perturbations: vary mass m, minimum branch separation d, superposition size Δx and interaction time t. Controls: shielding, charge neutralization, electromagnetic and Casimir–Polder forces, spin-pulse artifacts, vibration, and uncorrelated controls. Compare models at the same settings; identify regions where the quantum signal is resolvable and specified classical signals are not.

Candidate B | Metric diffusion versus loss of coherence

The question: Must a model that keeps spacetime classical predict a relationship between additional fluctuations and loss of matter-wave coherence? Observable families can include displacement or strain-noise spectra and coherence-loss bounds. These need not be measurable on one apparatus. Investigate whether jointly available observations already rule out parameter ranges. Recent 2026 work derives strain spectra for several classical-quantum models. [S03, S09]

Candidate C | Schrödinger–Newton optical cross-correlations

A January 2026 Physical Review D proposal uses a Michelson-type interferometer to test correlations between common and differential mirror motion expected under a specific semiclassical evolution. Reported feasibility is a model calculation with stated cryogenic and squeezed-light assumptions, not a demonstrated experiment or universal classical-gravity exclusion. [S10]

Experimental feasibility reality check. The 2025 EU-supported CLOSEtoQG project report describes why preparing picogram-scale masses in micrometre-scale spatial superpositions with roughly second-scale coherence is still an enormous challenge. Our first eight weeks are a theoretical and computational program—not an assertion that we can build this apparatus or collect new physical data. [S11]

Two 2026 alternatives belong in the experiment scan: paired atom interferometers that avoid some highly nonclassical massive states [S12], and mechanically constrained motions that may relax free-fall requirements under stated short-time assumptions [S13]. These proposals could change which apparatus offers the strongest near-term route.

Quantitative discrimination rule. For every candidate, specify the entire predicted measurement distribution P(y | model, parameters, nuisance noise), not just a slogan or a single expected mean. A hypothetical difference of 0.10 versus 0.13 in a measured correlation would not

discriminate models if the uncertainty is 0.20; a large separation compared with validated noise might. Optimize worst-case model separation over credible nuisance-parameter ranges, report the models that remain degenerate, and preregister design criteria before searching for favorable settings.
Selection gate. Score only after reproduction against identical criteria: (i) distinct predictions relative to known alternatives; (ii) physical feasibility/noise controls; (iii) mathematical tractability; (iv) proximity to existing apparatus or public observations; (v) potential genuinely new result. No prior commitment to one candidate.

THE EVIDENCE AND INFERENCE LADDER

Level 0 — Mathematical exercise: an equation or simulation is internally consistent; it may still not describe nature.
Level 1 — Reproduction: published analytical and numerical results match within stated domains and uncertainty.
Level 2 — Discrimination: named models differ for a specified observable under a specified set of assumptions and parameter values.
Level 3 — Experimental proposal: the difference survives validated nuisance models, quantified feasibility and available measurement precision.
Level 4 — Empirical inference: observation rules out specific models or parameter ranges at a stated confidence/likelihood level.
Level 5 — Fundamental claim: inference that gravity has nonclassical degrees of freedom requires the additional assumptions to be explicit; none of Levels 0–4 alone establishes a complete theory unifying gravity and quantum mechanics.

WORKING ARCHITECTURE — ONE NARRATIVE, ONE AUDIT TRAIL

Google Docs: THIS atlas is the readable source of context. Keep a short synopsis, a glossary, diagrams, examples, model comparisons, a chronological learning narrative, and a decision log. Never allow a 100-page undifferentiated document: expand detailed papers as linked annexes when needed.
Code record: charter approved; a local Git repository now preserves Cycle 01 code, tests, figures, source/claim registries and an uncompiled Lean draft. The complete export is backed up privately in Drive and linked from the Cycle 01 tab. The intended GitHub remote is not yet created because the connected action set lacks repository creation. Do not confuse a local commit with a GitHub push.
Python: NumPy/SciPy/SymPy, unit tests, parameter sweeps, uncertainty propagation, plotted observable distributions and data provenance; use pinned environments and deterministic random seeds. Avoid claiming experimental precision from an idealized simulation.
Lean 4 + mathlib: formalize narrow exact statements first: a complex 2×2 pure-state coefficient matrix has zero determinant exactly when that state factors (with defined nonzero-state

conditions); derive the branch-phase entanglement condition. Do not promise a formal proof of all quantum field theory, general relativity or apparatus physics.
Review protocol: one solver drafts; a distinct adversarial pass searches for counterexamples, missing terms, invalid limiting cases, dimensional inconsistencies and paper mismatches; computational tests and/or formal checks verify specifics. “Astra agrees with itself” is not evidence.
Provenance protocol: all scientific claims get a persistent ID, primary literature citations, classification (established/reproduced/proposed/disputed/open), date, theory assumptions, linked calculation or source, and explicit known objections. Negative results stay recorded.
The user-facing learning ritual. Every working session begins with a five-minute “what changed and why it matters” explanation; then one worked example with each symbol defined; then one diagram; then the strongest objection; then a choice or question for the research director. A question is welcome at any depth.

EIGHT-WEEK PROPOSED PROGRAM — AFTER APPROVAL

Gate 0 | APPROVED on 17 October 2026. Private work product; near-term priorities are verified calculations, understandable explanations and a credible discriminator. Cycle 01 reproductions are recorded in the first tab. The remaining scientific gates have not been waived.
Weeks 1–2 | Evidence baseline: acquire and closely read the primary model papers, extract equations/assumptions, build a comparison matrix and literature challenge register. First deliverable: teaching atlas v1 and three candidate comparison memos.
Weeks 3–4 | Reproduction: reproduce the ideal two-mass phase calculation, an established entanglement witness, at least one classical-channel/decoherence result, and one quantitative classical-entanglement or metric-diffusion result. First Lean algebra theorem. Independent numerical checks required.
Weeks 5–6 | Discriminator search: quantify where predictions diverge with nuisance parameters marginalized or bounded; explore sensitivity to m, d, Δx and t, alternative observables, and whether a two-experiment joint constraint is materially stronger. Register candidate innovations and aggressively try to falsify each.
Weeks 7–8 | Stress test and draft: assess instrument feasibility, model completeness, likelihood or information-discrimination metric, counterexamples and published priority. Produce a reproducible technical note. Seek qualified specialist review before public claims or submission.
At every gate: either CONTINUE with evidence, PIVOT with a documented reason, or STOP because the purported novelty already exists or the predicted signal cannot survive realistic constraints.

DELIVERABLES AND REVIEW STANDARDS

After each research cycle: 1-page plain-language briefing; updated figure; new or corrected definitions; what calculations passed; what failed; strongest new counterargument; evidence

links; next decision. Code must execute cleanly from a pinned environment. A public claim needs independent physical review.
Success minimum: a correct public, reproducible map of models, assumptions and observables plus verified toy calculations. Strong success: a previously unreported, independently checked parameter bound or experiment-design insight. Exceptional success: a publishable, new discrimination result. Unification itself is beyond the scope of this program.

LEARNING CHECKPOINT — CAN YOU ANSWER THESE?

Why can two particles be correlated without being entangled? What does the phase Δφ contain that a common phase does not? Why does a classical field not necessarily imply a purely LOCC information channel? What combination of measurements would reject a specified model even when it predicts some entanglement?
These are teaching prompts, not prerequisites. Each will acquire an interactive example or diagram during the research program.

PRIMARY SOURCES — REGISTRY v0.1

S01 Bose et al. (2017). Spin entanglement witness of quantum gravity. Physical Review Letters 119, 240401. https://doi.org/10.1103/PhysRevLett.119.240401
S02 Kafri, Taylor & Milburn (2014). A classical channel model for gravitational decoherence. New Journal of Physics 16, 065020. https://doi.org/10.1088/1367-2630/16/6/065020
S03 Oppenheim (2023). A postquantum theory of classical gravity? Physical Review X 13, 041040. https://doi.org/10.1103/PhysRevX.13.041040
S04 Aziz & Howl (2025). Classical theories of gravity produce entanglement. Nature 646, 813–817. https://doi.org/10.1038/s41586-025-09595-7
S05 Trillo & Navascués (2025). Diósi–Penrose model of classical gravity predicts gravitationally induced entanglement. Physical Review D 111, L121101. https://doi.org/10.1103/PhysRevD.111.L121101
S06 Di Biagio (accepted 2026). Gravity-induced entanglement is not a theory-independent witness of nonclassicality of the gravitational field. Physical Review D. https://doi.org/10.1103/r8ry-sp35
S07 Boulle & Franzmann (2026). Subsystems independence in gravitationally induced entanglement proposals. Physical Review D 114, 025015. https://doi.org/10.1103/tgjx-y7hl
S08 Feng, Vedral & Marletto (2026). Collapse-based models for gravity do not violate the entanglement-based witness of nonclassicality. Physical Review D 113, 104055. https://doi.org/10.1103/83rl-nygv
S09 Hirotani & Matsumura (2026). Testing classical-quantum gravity with geodesic deviation. Physical Review D 114, 026014. https://doi.org/10.1103/fx1h-97sx
S10 Liu, Chen, Somiya & Ma (2026). Testing the quantum nature of gravity through interferometry. Physical Review D 113, 022002. https://doi.org/10.1103/15n3-zzjw

S11 European Commission / CORDIS (last updated February 2025). CLOSEtoQG experimental-platform progress and constraints. https://cordis.europa.eu/project/id/101041115/reporting
S12 Howl, Cooper & Hackermüller (2026). Gravitationally induced entanglement in atom interferometry. Physical Review A 114, 023306. https://doi.org/10.1103/l62d-gz5c
S13 Williams (2026). Gravity-induced entanglement under constrained dynamics. Physical Review A 114, 032221. https://doi.org/10.1103/pg7l-f26t
Source policy: the registry is an initial reading list, not a claim that every equation has been independently audited yet. Primary published papers outrank summaries. Flag contradictory claims explicitly instead of silently harmonizing them.

DECISION LOG / CHANGELOG

2026-10-17 | v0.1 charter created. Scope changed from “entanglement is a proof” to a comparative, multi-observable discrimination program. Three experiment families nominated; none selected. Model registry and epistemic ladder established. Research execution remains on hold pending approval.